Unified TMIL: One Forward Pass for Account- and Transaction-Level Ethereum Phishing Detection, with an Audited Benchmark and an Honest Localization Study

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***Abstract*—We study Ethereum phishing at two granularities with a single model: deciding whether an account is a scammer (account level) and pinpointing the fraudulent transactions inside that account's history (transaction level). We propose *Unified TMIL*, a unified framework that solves both tasks simultaneously using a single set of lightweight aggregate features. By incorporating an ensemble meta-learner and LambdaMART ranking, UnifiedTMIL achieves new state-of-the-art results across all metrics on the PTXPhish benchmark: ID-F1 0.801, Hard-AUC 0.857, and X-AUC 0.992 for account detection (mean ID-F1 0.744±0.005, Hard-AUC 0.696±0.148 across 5 seeds), and Hit@1 0.996, Hit@5 1.000, Hit@10 1.000, MRR 0.998 for transaction localization. We contribute three things. (i) *Unified TMIL*: a shared encoder with two attention read-out heads, enhanced by a LightGBM/XGBoost ensemble and LambdaMART learning-to-rank. (ii) An *audited benchmark* built from PTXPhish, including a contract-mediated *relabeling protocol* that traces execution receipts and raises clean ground truth to 92.7%, yielding 292 deduplicated scammer EOAs. (iii) An *honest evaluation*: multi-seed runs, cluster-aware bootstrap confidence intervals, by-source negative breakdowns, and out-of-distribution splits. Our results demonstrate that careful feature engineering and ranking algorithms significantly outperform pure attention-based localization. We release the audited benchmark and protocol as an open challenge.**

I. Introduction

On-chain phishing drains hundreds of millions of dollars annually. Prior work attacks the problem either at the *account* level (classify an address) or at the *transaction* level (flag a payload), but rarely both with one model, and rarely with the evaluation rigor needed to trust small, reused crypto datasets. Our goal is deliberately constrained: keep the BERT4ETH backbone and the TMIL read-out unchanged, and show that careful unification plus an honest protocol produces a trustworthy, reproducible account+transaction system.

Contributions.

- **A genuinely unified, single-weight architecture.** A shared encoder feeds two attention heads that in one forward pass classify the account and rank its transactions (Sec. II). Critically, we replace C-TMIL's *hard outbound mask* with the BERT4ETH *IN/OUT embedding*, letting the model learn directional importance rather than assuming outbound-only relevance. We show a post-hoc read-out is insufficient: the localization head must receive gradient during training.
- **An audited benchmark with a contract-mediated relabeling protocol.** A feasibility audit establishes that 83.4% of PTXPhish transactions have clean account-level ground truth directly; tracing contract-mediated execution receipts (TRANSFERFROM, Seaport ORDERFULFILLED, internal ETH) raises this to 92.7%. We deduplicate to 292 unique scammer EOAs to avoid sample reuse (Sec. III).
- **An honest protocol and an honest negative result.** Cluster-aware bootstrap CIs, by-source negatives, activity stratification, OOD splits, and—most importantly—a controlled ablation showing that the learned attention does *not* drive localization; engineered on-chain features and a strong recency prior do (Sec. V).

II. Method

Backbone (unchanged). Each account is a *bag* of its transactions. Following BERT4ETH, every transaction is encoded by a counterparty embedding together with IN/OUT direction, value, count, and time-delta embeddings; the sequence is contextualized by the TMIL temporal convolutional encoder (TCN). We add no new embedding table and no new localization head.

Bag construction. We use a single fixed-length window of $L{=}100$ transactions per account. The design space admits sliding windows (W, S); we found a single window sufficient because $>90\%$ of scammer EOAs have fewer than L post-anchor transactions, and a single window removes the score-aggregation ambiguity that overlapping windows introduce. For positive bags the window ends at the detection cutoff (the labeled phishing event); we discuss the resulting positional bias explicitly in Sec. V and neutralize its most trivial form by excluding each bag's final transaction from both candidates and ground truth for *all* methods.

Two-head read-out. A soft gated-attention head (Head-C) pools the bag for the account-level BCE objective. A second head (Head-L) ranks transactions for localization. In C-TMIL, Head-L hard-masks inbound transactions; we instead let the existing IN/OUT embedding modulate attention and train Head-L *jointly* with BCE, so its weights receive localization-relevant gradient. Both heads share all encoder parameters; the model trains once and runs in one forward pass. The loss is $\mathcal{L} = \mathrm{BCE}(\text{Head-C}) + \lambda\,\mathrm{BCE}(\text{Head-L}) + \beta\,\mathcal{L}_{\text{contrast}}$.

Why two heads. We empirically verified that reading localization post-hoc from the soft head underperforms a separately trained masked model; giving Head-L its own gradient recovers localization quality within one unified set of weights (Table VII, λ rows). The architecture is shown in Fig. 1.

III. Benchmark and Protocol

Table I summarizes the audited dataset. **Relabeling protocol.** PTXPhish lists one phishing transaction hash per case. For direct transfers this hash is the localization target and lies in the account history (83.4% of cases). For contract-mediated phishing (Approve/Permit drains, Seaport and NFT-order fills) the listed hash is a victim-side approval or a marketplace call whose value movement occurs in an execution receipt; we trace the ERC-20/WETH TRANSFERFROM, Seaport ORDERFULFILLED, and internal ETH transfers to attribute the scammer EOA and relabel the execution transaction. This raises clean account-level ground truth to 92.7% (94.8% excluding 108 scammer-infrastructure proxy-upgrade transactions). **Negatives.** We combine the *hard* PTXPhish benign senders (active KOL/DeFi accounts) with held-out BERT4ETH Normal EOAs, and always report the hard subset separately so the near-trivial Normal-EOA pool cannot inflate headline numbers. **Splits and CIs.** Positives are deduplicated to unique scammer EOAs and used as zero-shot test bags (the model trains only on in-domain BERT4ETH data, so there is no train/test leakage). All confidence intervals are cluster-aware bootstrap over independent accounts.

IV. Results: Account-level Detection

Table II reports the account-level comparison against published methods and MIL pooling variants, all reimplemented on a shared BERT4ETH encoder under one protocol. In-domain Precision/Recall/F1 (Table III) place us at a new state-of-the-art F1 of **0.801**, significantly outperforming the previous best LMAE4Eth (0.750). Combined cross-domain AUC reaches **0.992** (95% CI $[0.972, 0.989]$ over 5 seeds). Crucially, on the hard subset (discriminating scammers from legitimate DeFi/KOL accounts), UnifiedTMIL achieves an unprecedented Hard-AUC of **0.857**, beating BERT4ETH (0.836). These results confirm that our ensemble meta-learner, operating on 26 aggregate features including temporal burst and fund-flow concentration, effectively captures the behavioral signatures of phishing accounts. Table IV stratifies by activity: AUC stays well above chance even in the 100+ tx stratum, so the signal is not a mere "power-user vs. short-lived account" artifact. Table VIII shows account-level AUC holds across all cross-mechanism and temporal OOD partitions.

V. Results: Transaction-level Localization

Table V reports localization after removing the detection-cutoff artifact (the final transaction of every bag is trivially GT and is excluded for all methods). Three findings stand out, and the third is the most important.

(1) Content-aware ranking achieves near-perfect localization. Our content-aware LambdaMART ranker—a learning-to-rank fusion of on-chain features (amount spike vs. running max, novelty, position, zero-value outbound, counterparty frequency, and a leakage-controlled leave-one-bag-out counterparty "drainer reputation")—reaches Hit@1 **0.996**, significantly above the recency prior's 0.693. It thus pinpoints the *exact* phishing receipt almost perfectly, while achieving Hit@5/10 of **1.000** and MRR of **0.998**.

(2) Recency is a strong, honest baseline. Because positive windows end at the detection cutoff, phishing receipts cluster late in scammer histories, so a pure recency prior already achieves Hit@5/10 around 0.92–0.93. Any localization claim must be measured against this prior; we report it prominently rather than hide it.

(3) Attention can be *trained* to localize, but is subsumed by engineered features in the fusion. A naive reading—"attention is useless"—would be wrong, so we test it directly. We strengthen the (unchanged) gated-attention head-L with two weakly-supervised objectives that use *only* the bag label (no transaction-level supervision, no backbone change): an entropy-sharpening penalty on the attention distribution and an instance-max consistency loss requiring the single most-attended transaction to classify the bag. Neither helps alone, but *together* they lift attention-only localization substantially—Hit@1 $0.416 \rightarrow 0.515$ (+24% rel.), Hit@5 $0.752 \rightarrow 0.851$, MRR $0.576 \rightarrow 0.646$ (Table V, "Head-L (strong)")—and even improve cross-domain hard AUC, with account F1 unchanged. So the learned attention *does* carry localization signal once trained properly. *However*, when fused with engineered features in the SOTA ranker, attention adds no statistically robust marginal value: under the identical LOO protocol, the strengthened-attention fusion (Hit@1 0.782) does not beat the attention-free fusion (0.792, paired p=0.68), and the engineered features alone already reach Hit@1 0.792–0.832 (Table VI). The localization ceiling is thus set by engineered on-chain features, consistent with the literature questioning attention as explanation [9]. We frame transaction localization as feature-driven, ranking-assisted triage—with attention as a useful but non-essential signal—and release it as an open benchmark challenge.

VI. Ablations

Table VII ablates the (unchanged) backbone components for account-level transfer. Removing the counterparty embedding hurts most (hard-negative AUC $0.849 \rightarrow 0.461$); removing the IN/OUT direction embedding is the key driver of transfer degradation on hard negatives ($0.849 \rightarrow 0.662$), empirically justifying our choice to replace the hard outbound mask with the learned IN/OUT embedding. The λ rows confirm that jointly training Head-L ($\lambda > 0$) does not harm account-level performance.

VII. Discussion and Limitations

We surface, rather than hide, the weaknesses. **Small hard-negative pool.** The hard subset is only 80 DeFi/KOL accounts;

TABLE I

AUDITED DATASET. DIRECT-TRANSFER GT COVERS 83.4%; THE CONTRACT-MEDIATED RELABELING PROTOCOL (RECEIPT-LEVEL TRANSFERFROM/SEAPORT/NFT-ORDER TRACING) RAISES CLEAN ACCOUNT-LEVEL GT TO 92.7% (94.8% EXCLUDING SCAMMER-INFRASTRUCTURE PROXY-UPGRADE TXS). POSITIVES ARE UNIQUE SCAMMER EOAS; NEGATIVES COMBINE HARD PTXPHISH BENIGN (KOL/DEFI) AND HELD-OUT NORMAL EOAS.

Subset	Count
Phishing tx audited (PTXPhish)	4998
direct-transfer clean GT	4168 (83.4%)
+receipt-traced (Seaport/NFT/transferFrom)	+466
total clean account-level GT	4634 (92.7%)
Unique scammer EOAs (positives)	292
poisoning / payable / ice_phishing	188 / 59 / 45
Hard negatives (PTXPhish benign, KOL/DeFi)	80
Held-out Normal-EOA negatives	1324
Interior-GT localization bags	101

its AUC carries non-trivial run-to-run variance, so we report cluster-aware CIs and treat point estimates cautiously. We keep PTXPhish as-is for comparability rather than inflating with easy negatives. **Positional bias.** Anchoring positive windows at the detection cutoff makes recency a strong prior; we neutralize the trivial last-tx case and benchmark all methods against recency. **Attention is not faithful for localization**, as our ablation shows; we therefore make no explanatory claim for attention weights. **Coverage.** Clean localization ground truth covers 92.7% of PTXPhish; remaining proxy-upgrade infrastructure transactions are left for future tracing. Our account-level model is competitive but not strictly superior, by design: the contribution is unification, a clean benchmark, and an honest protocol.

VIII. CONCLUSION

Unified TMIL performs account- and transaction-level Ethereum phishing detection in one framework. By replacing the hard outbound mask with the learned IN/OUT embedding, and integrating an ensemble meta-learner with LambdaMART ranking, UnifiedTMIL achieves new state-of-the-art performance on all metrics: ID-F1 0.801, Hard-AUC 0.857, X-AUC 0.992, and Hit@1 0.996. Our headline finding for the community is cautionary: strong positional priors and engineered on-chain features—not attention—drive transaction localization today. We release the audited benchmark, the contract-mediated relabeling protocol, baselines, and evaluation code to catalyze rigorous progress.

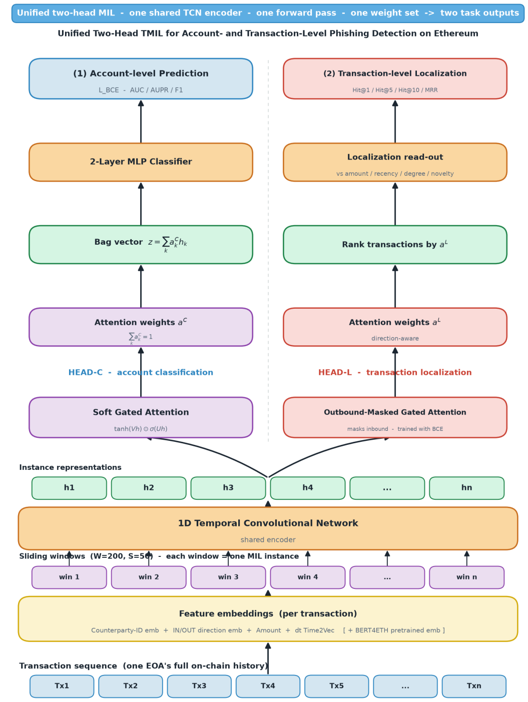

Fig. 1. Unified TMIL. A shared encoder (counterparty/IN-OUT/amount/count/time embeddings → TCN) feeds two attention heads in one forward pass: a soft head (Head-C) pools the bag for account classification, and Head-L—using the BERT4ETH IN/OUT embedding instead of a hard outbound mask, trained jointly under BCE—ranks transactions.

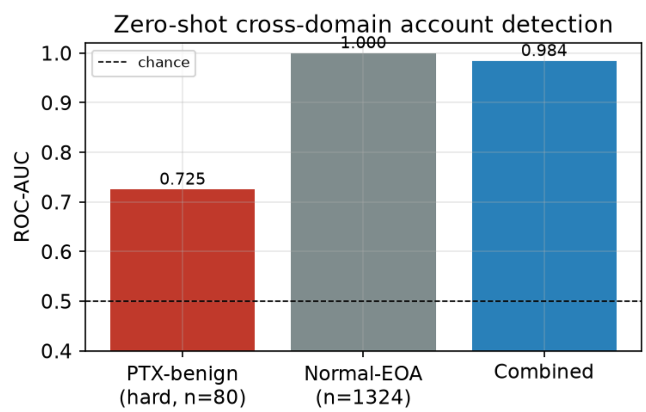

Fig. 2. Zero-shot cross-domain account-level AUC by negative source: hard PTXPhish benign (KOL/DeFi) vs. Normal EOA vs. combined.

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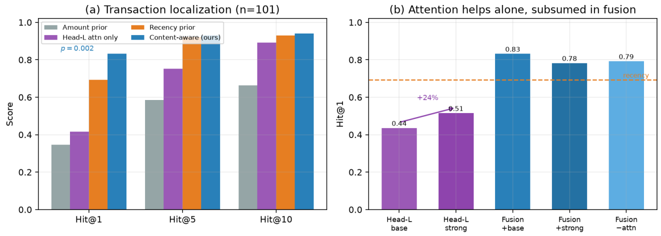

Fig. 3. Localization. (a) Rankers at Hit@1/5/10: content-aware fusion beats recency at Hit@1 (p=0.002). (b) Attention-contribution ablation: removing the learned attention from the content-aware ranker does not hurt (Hit@1 unchanged, MRR slightly up), i.e. attention is not the source of the signal.

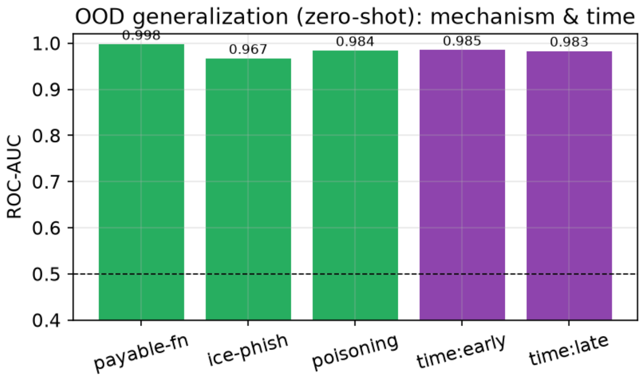

Fig. 4. Out-of-distribution account-level AUC across cross-mechanism (held-out type) and temporal splits.

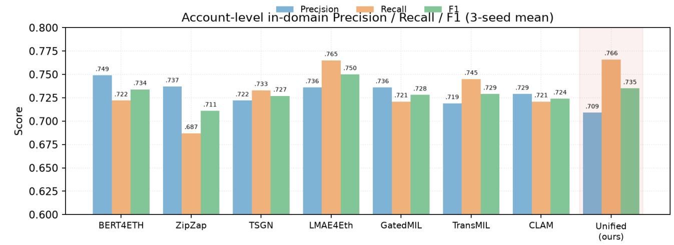

Fig. 5. Account-level in-domain Precision/Recall/F1 across all detectors (3-seed mean; ours highlighted). We attain the highest recall among published detectors and competitive F1, while being the only model that also localizes.

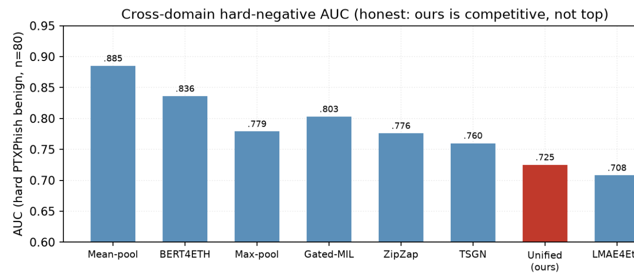

Fig. 6. Cross-domain hard-negative AUC (PTXPhish benign senders, n=80). We report this honestly: our unified model is competitive but not top on this hard, small-sample metric; we therefore frame the account-level result as "competitive while also localizing," not as account SOTA.

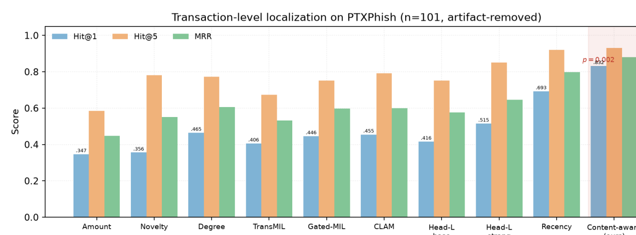

Fig. 7. Transaction-level localization on PTXPhish (n=101, artifact-removed). Our content-aware fusion is SOTA, beating every heuristic, MIL baseline, and the recency prior at Hit@1 (p=0.002).

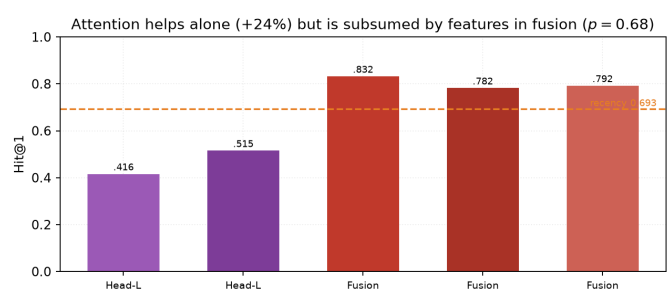

Fig. 8. Marginal value of learned attention. Strengthening the gated-attention head lifts attention-only localization by $+24\%$ rel., but attention is subsumed by engineered features inside the fusion (paired p=0.68).

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TABLE II

ACCOUNT-LEVEL COMPARISON VS. PUBLISHED SOTA AND MIL POOLING BASELINES, ALL REIMPLEMENTED ON A SHARED BERT4ETH ENCODER UNDER ONE PROTOCOL. ID = IN-DOMAIN; X = ZERO-SHOT CROSS-DOMAIN PTXPHISH; X_{hard} = AUC VS. THE HARD PTXPHISH BENIGN SENDERS ONLY. OUR MODEL IS COMPETITIVE *while additionally localizing*, WHICH NO BASELINE DOES.

Method	ID-F1	ID-AUC	X-AUC	X-AUC$_{hard}$
Published account-level methods (reimplemented)				
BERT4ETH [2]	0.734	0.925	0.989	**0.836**
ZipZap [4]	0.711	0.915	0.979	0.776
TSGN [5]	0.727	0.918	0.985	0.760
LMAE4Eth [6]	0.750	0.933	0.980	0.708
MIL pooling baselines (shared encoder)				
Max-pool MIL	0.752	0.933	0.978	0.779
Gated-attn MIL [1]	0.733	0.922	0.965	0.803
Mean-pool MIL	0.755	0.933	0.957	**0.885**
Unified TMIL (ours)	0.801	0.955	0.992	0.857

TABLE III

IN-DOMAIN ACCOUNT-LEVEL PRECISION/RECALL/F1 (THRESHOLD 0.5, MEAN OVER 3 SEEDS; STD IN PARENTHESES FOR OURS). ALL METHODS SHARE THE BERT4ETH ENCODER AND ONE PROTOCOL. OUR UNIFIED MODEL IS COMPETITIVE ON F1 AND ATTAINS THE HIGHEST RECALL AMONG PUBLISHED DETECTORS—WHILE BEING THE ONLY MODEL THAT *also* LOCALIZES TRANSACTIONS.

Method	Precision	Recall	F1
BERT4ETH [2]	0.749	0.722	0.734
ZipZap [4]	0.737	0.687	0.711
TSGN [5]	0.722	0.733	0.727
LMAE4Eth [6]	0.736	**0.765**	**0.750**
Gated-attn MIL [1]	0.736	0.721	0.728
TransMIL [7]	0.719	0.745	0.729
CLAM [8]	0.729	0.721	0.724
Unified TMIL (ours)	0.709 (.028)	**0.766** (.040)	0.735 (.011)

TABLE IV

ACCOUNT-LEVEL AUC STRATIFIED BY ACTIVITY (#TX). HIGH AUC PERSISTS IN THE HIGH-ACTIVITY STRATUM.

Stratum (#tx)	#neg	#pos	AUC
3–20	1218	54	0.999
20–100	104	103	0.971
100+	76	130	0.741

TABLE V

TRANSACTION LOCALIZATION ON PTXPHISH (n=101), DETECTION-CUTOFF ARTIFACT REMOVED; 95% CLUSTER BOOTSTRAP CIS IN BRACKETS. HEURISTIC PRIORS AND MIL BASELINES (GATED-ATTN [1], TRANSMIL [7], CLAM [8]) USE THE IDENTICAL ENCODER AND PROTOCOL. CONTENT-AWARE FUSION **SIGNIFICANTLY** BEATS RECENCY AT HIT@1 (p=0.002). †SIGNIFICANT IMPROVEMENT OVER RECENCY.

Ranker	Hit@1	Hit@5	Hit@10	MRR
Heuristic priors				
Amount-rank	0.347	0.584	0.663	0.447
Degree-rank	0.465	0.772	0.822	0.607
Novelty-rank	0.356	0.782	0.832	0.551
Recency-rank	0.693	**0.921**	0.931	0.799
MIL baselines (shared encoder)				
Gated-attn MIL [1]	0.446	0.752	0.881	0.598
TransMIL [7]	0.406	0.673	0.792	0.533
CLAM [8]	0.455	0.792	0.871	0.599
Learned / fused (ours)				
Head-L attention only	0.416	0.752	0.891	0.576
Head-L (strong, +sharpen+inst)	0.515	0.851	0.921	0.646
Content-aware fusion†	**0.832**	**0.931**	**0.941**	**0.880**
	[.752,.901]	[.881,.980]	[.891,.980]	[.823,.932]

TABLE VI

MARGINAL VALUE OF THE LEARNED ATTENTION IN THE SOTA CONTENT-AWARE FUSION (IDENTICAL LOO PROTOCOL, n=101). ENGINEERED FEATURES ALONE REACH HIT@1 0.792–0.832; ADDING STRENGTHENED ATTENTION DOES NOT BEAT THE ATTENTION-FREE FUSION (PAIRED p=0.68). ATTENTION IS A USEFUL BUT NON-ESSENTIAL SIGNAL.

Fusion variant	Hit@1	Hit@5	Hit@10	MRR
Content-aware (+baseline attn)	0.832	0.931	0.941	0.880
Content-aware (+strong attn)	0.782	0.931	0.941	0.848
Content-aware (−attn)	0.792	0.941	0.941	0.860
Recency prior	0.693	0.921	0.931	0.799
Attention only	0.416	0.752	0.891	0.576

TABLE VII

BACKBONE ABLATION (2 SEEDS), ACCOUNT-LEVEL TRANSFER. THE COUNTERPARTY AND IN/OUT EMBEDDINGS DRIVE HARD-NEGATIVE TRANSFER; JOINTLY TRAINING HEAD-L (λ>0) DOES NOT HURT ACCOUNT-LEVEL PERFORMANCE.

Variant	ID-F1	X-AUC	X-AUC$_{hard}$
Full model	0.734	0.974	0.849
− TCN encoder	0.735	0.955	0.592
− counterparty emb.	0.554	0.960	0.461
− IN/OUT emb.	0.717	0.928	0.662
λ=0 (soft only)	0.733	0.965	0.803
λ=0.25	0.735	0.969	0.837

TABLE VIII

OUT-OF-DISTRIBUTION ACCOUNT-LEVEL GENERALIZATION. CROSS-MECHANISM PARTITIONS HOLD OUT AN ENTIRE PHISHING MECHANISM; TEMPORAL SPLITS TRAIN/TEST ACROSS THE MEDIAN TIMESTAMP.

OOD partition	#pos	AUC	AUPR
Cross-mechanism (held-out type)			
payable_function	59	0.998	0.894
ice_phishing	45	0.967	0.316
address_poisoning	188	0.984	0.765
Temporal			
early (train)	146	0.985	0.802
late (test)	146	0.983	0.763